

Dominick Reilly

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Education

University of North Carolina at Charlotte

Aug 2022 - Present

Ph.D. Student in Computer Science

Advisor: Dr. Srijan Das

University of North Carolina at Charlotte

Jan 2019 - May 2022

Bachelor of Science in Computer Science, GPA: 4.0

Experience

University of North Carolina at Charlotte

Aug 2022 - Present

Research Assistant - Charlotte Machine Learning Lab

- Domain adaption of Vision Language Models (VLMs) through visual probing, enabling adaptation to novel visual domains and preventing catastrophic forgetting of source domains. [1]
- Cross-view (egocentric → exocentric) knowledge transfer in VLMs for enhancing egocentric understanding in exocentric videos. [6, 2]
- Multi-modal (RGB + Human Pose + Human Object Interactions) Large Language Models (LLM) for understanding daily living actions from video. Our approach leverages a curriculum-based training strategy for integrating new modalities into the LLM embedding space. [4]
- Multi-modal (RGB + Human Pose) and viewpoint agnostic Vision Transformers (ViT) for understanding subtle appearance and motion cues in human activity videos. [10]
- Self-supervised learning and masked autoencoders for training ViTs on small, out-of-domain data distributions. [11]

Honda Research Institute, California

Jun 2024 - Sep 2024

Student Researcher - Perception Team

- View-invariant VLMs for procedural video understanding. View-invariant representations are learned on cross-view videos (ego & exo) and improve performance on single-view videos.
- Learning pose-augmented RGB representations in CNNs for weakly-supervised action and error detection in procedural activity videos.

Inria, Sophia Antipolis, France

Jan 2024 - Jun 2024

Research Intern - STARS Team (Chateaubriand Fellow)

- Multi-modal (RGB + Human Pose) CLIP models for improved zero-shot understanding of daily living activities from video. [8] [5]
- Learnable gating mechanisms to improve feature selection for temporal activity detection.[9]

University of North Carolina at Charlotte

Jul 2021 - Jul 2022

Research Assistant - Data Privacy Lab

- Safeguarding face and iris images from deep-learning based re-identification attacks. [13]
- Created interactive webpage demonstrating safeguards on face and iris images. Try the demo for yourself at <http://3.223.148.187/>. [12]

Publications (C)onference, (P)reprint, (W)orkshop

1. (P) **Dominick Reilly**, Manish Kumar Govind, Le Xue, Srijan Das, “ViSCoP: Visual Probing for Video Domain Adaptation of Vision Language Models,” Under Review (PDF available on my webpage), 2025.
2. (W) **Dominick Reilly**, Manish Kumar Govind, Le Xue, Srijan Das, “From My View to Yours: Ego-Augmented Learning in Large Vision Language Models for Understanding Exocentric Daily Living Activities,” MMFM3: The 3rd Workshop on What is Next in Multimodal Foundation Models? at CVPR, 2025.
3. (W) **Dominick Reilly**, Srijita Das, Srijan Das, “Learning Pose-aware Representations in Vision Transformers for Understanding Activities of Daily Living,” 8th Workshop and Competition on Affective & Behavior Analysis in-the-wild at CVPR, 2025.
4. (C) **Dominick Reilly**, Rajatsubhra Chakraborty, Arkaprava Sinha, Manish Kumar Govind, Pu Wang, Francois Bremond, Le Xue, Srijan Das, “LLAVIDAL: A Large Language Vision Model for Daily Activities of Living,” IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025.
5. (C) Arkaprava Sinha, **Dominick Reilly**, Francois Bremond, Pu Wang, Srijan Das, “SKI Models: Skeleton Induced Vision-Language Embeddings for Understanding Activities of Daily Living,” Proceedings of the AAAI Conference on Artificial Intelligence (AAAI), 2025.
6. (P) **Dominick Reilly**, Manish Kumar Govind, Le Xue, Srijan Das, “From My View to Yours: Ego-Augmented Learning in Large Vision Language Models for Understanding Exocentric Daily Living Activities,” arXiv:2501.05711, 2024.
7. (P) Ali Khaleghi Rahimian, Manish Kumar Govind, Subhajit Maity, **Dominick Reilly**, Christian Kümmerle, Srijan Das, Aritra Dutta, “Fibottention: Inceptive Visual Representation Learning with Diverse Attention Across Heads,” arXiv:2406.19391, 2024.
8. (W) Mahmoud Ali, Di Yang, Arkaprava Sinha, **Dominick Reilly**, Srijan Das, Gianpiero Francesca, Francois Bremond, “Quo Vadis, Video Understanding with Vision-Language Foundation Models?,” Workshop on Video-Language Models (NeurIPS), 2024.
9. (W) Aglind Reka, Diana Laura Borza, **Dominick Reilly**, Michal Balazsia, Francois Bremond, “Introducing Gating and Context into Temporal Action Detection,” Workshop on Affective Behavior Analysis in-the-wild at (ECCV), 2024.
10. (C) **Dominick Reilly** and Srijan Das, “Just Add π ! Pose Induced Video Transformers for Understanding Activities of Daily Living,” IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024.
11. (C) Srijan Das, Tanmay Jain, **Dominick Reilly**, Pranav Balaji, Soumyajit Karmakar, Shyam Marjit, Xiang Li, Abhijit Das, Michael Ryoo, “Limited Data, Unlimited Potential: A Study on ViTs Augmented by Masked Autoencoders,” IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2024.
12. (C) Muhammad Saleem, **Dominick Reilly**, Liyue Fan, “DP-Shield: Face Obfuscation with Differential Privacy,” International Conference on Extending Database Technology, 2022.
13. (C) **Dominick Reilly**, Liyue Fan, “Comparative Evaluation for Differentially Private Image Obfuscation,” IEEE International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications (IEEE TPS), 2021.

Patents filed

1. Srijan Das and **Dominick Reilly**. Large Language Vision Model (LLVM), US Patent Application No. 63/693,982 filed on Sep 12, 2024.

Academic Activities

- Reviewer for CVPR 2025, AAAI 2023/24/25

Awards & Achievements

1. CVPR 2025 Travel Award
2. The Chateaubriand Fellowship (awarded by the Embassy of France), 2023
3. Best poster award in Mathematics and Computer Science, UNC Charlotte Undergraduate Research Conference, 2020